Lecture 2. Axioms of Probability

Sample spaces, axioms, inclusion–exclusion, equally likely outcomes, continuity of probability

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2.1 Sample space and events

The **sample space** S is the set of all possible outcomes of an experiment; an **event** is a subset of S. For events E, F we use the union $E \cup F$, the intersection $EF = E \cap F$, the complement $E^c = S \setminus E$, and the empty event $\emptyset$. E and F are **mutually exclusive** if $EF = \emptyset$. Union and intersection are commutative, associative and distribute over each other, and

$$\Big(\bigcup_{i=1}^{n} E_i\Big)^c = \bigcap_{i=1}^{n} E_i^c, \qquad \Big(\bigcap_{i=1}^{n} E_i\Big)^c = \bigcup_{i=1}^{n} E_i^c \qquad \text{(De Morgan)}.$$

For the first, an outcome lies outside every E_i exactly when it lies outside their union; the second follows by applying the first to the E_i^c.

2.2 Axioms of probability

The relative-frequency idea, $P(E) = \lim n(E)/n$ over repeated trials, motivates but does not define probability; instead we impose axioms.

Definition (Probability)

A function P on events is a **probability** if (i) $0 \le P(E) \le 1$;
(ii) $P(S) = 1$;
(iii) for mutually exclusive $E_1, E_2, \ldots$, $P\Big(\bigcup_{i=1}^{\infty} E_i\Big) = \sum_{i=1}^{\infty} P(E_i)$.

Taking $E_1 = S$ and $E_i = \emptyset$ for $i \ge 2$ forces $P(\emptyset) = 0$, and then (iii) gives finite additivity. (For uncountable S one can only define P on a suitable class of events; we ignore this.)

Proposition 2.1

(a) $P(E^c) = 1 - P(E)$.
(b) $E \subset F \Rightarrow P(E) \le P(F)$.
(c) $P(E \cup F) = P(E) + P(F) - P(EF)$.

Proof. (a) E and E^c are disjoint with union S.

(b) $F = E \cup E^c F$, a disjoint union.

(c) $E \cup F = E \cup E^c F$ and $F = EF \cup E^c F$, both disjoint; subtract. ∎

Theorem 2.2 (Inclusion–exclusion)

$$P\Big(\bigcup_{i=1}^{n} E_i\Big) = \sum_i P(E_i) - \sum_{i<j} P(E_i E_j) + \ldots + (-1)^{r+1} \sum_{i_1<\cdots<i_r} P(E_{i_1} \cdots E_{i_r}) + \ldots + (-1)^{n+1} P(E_1 \cdots E_n).$$

Proof. For a finite or countable S it suffices to check that each outcome contributes its probability equally to both sides. An outcome in none of the E_i contributes 0 to both. An outcome in

exactly $m \geq 1$ of them contributes once on the left, and on the right $\binom{m}{1} - \binom{m}{2} + \cdots \pm \binom{m}{m} = 1 - \sum_{i=0}^{m} (-1)^i \binom{m}{i} = 1$ times. (For general S, induct on n using Proposition 2.1(c).) ■

Truncating the sum gives alternately upper and lower bounds (Bonferroni): $P(\bigcup E_i) \leq \sum P(E_i)$ and $P(\bigcup E_i) \geq \sum P(E_i) - \sum_{i<j} P(E_i E_j)$, and so on.

2.3 Equally likely outcomes

If $S = \{1, \ldots, N\}$ and every outcome has probability $1/N$, then $P(E) = |E|/N$, and probability reduces to Lecture 1. It is often convenient to treat objects as distinguishable even when they are not; this never changes the answer and makes outcomes equally likely.

Example 2.1

(a) Three balls are drawn from an urn with 6 white and 5 black balls. Then $P(\text{one white, two black}) = \binom{6}{1}\binom{5}{2}/\binom{11}{3} = 60/165 = 4/11$. Counting ordered draws gives $3 \cdot 6 \cdot 5 \cdot 4/(11 \cdot 10 \cdot 9) = 4/11$ as well.
(b) k of n balls are drawn, one of them special. By symmetry, the special ball is equally likely to be in any position of a random ordering, so $P(\text{special drawn}) = k/n$; directly, $\binom{1}{1}\binom{n-1}{k-1}/\binom{n}{k} = k/n$.
(c) A poker full house: choose the rank of the triple and its 3 suits, then the rank of the pair and its 2 suits: $13\binom{4}{3} \cdot 12\binom{4}{2}/\binom{52}{5} = 3744/2{,}598{,}960 \approx 0.00144$.

Example 2.2 (Birthday problem)

Ignoring leap years and assuming all 365^n birthday assignments equally likely, the probability that n people have distinct birthdays is $365 \cdot 364 \cdots (366 - n)/365^n$. This drops below $1/2$ at $n = 23$: the probability of a shared birthday is 0.507 for $n = 23$ and 0.970 for $n = 50$. The surprise disappears when one counts pairs: 23 people form $\binom{23}{2} = 253$ pairs, each matching with probability $1/365$.

Example 2.3 (The card after the first ace)

A shuffled deck is turned up card by card. Is the card following the first ace more likely to be the ace of spades or the two of clubs? Count orderings in which the ace of spades follows the first ace: remove it, order the other 51 cards (51! ways), and reinsert it right after the first ace. So the probability is $51!/52! = 1/52$. The same argument applies to the two of clubs, so both have probability $1/52$.

Example 2.4 (The matching problem)

N people throw their hats into a pile and each picks one at random. Let E_i be the event that person i gets their own hat. For $i_1 < \cdots < i_r$, $P(E_{i_1} \cdots E_{i_r}) = (N - r)!/N!$, and there are $\binom{N}{r}$ such terms, so by inclusion–exclusion

$$P\Big(\bigcup_i E_i\Big) = \sum_{r=1}^{N} (-1)^{r+1} \binom{N}{r} \frac{(N-r)!}{N!} = \sum_{r=1}^{N} \frac{(-1)^{r+1}}{r!},$$

so that

$$P(\text{no match}) = \sum_{r=0}^{N} \frac{(-1)^r}{r!} \xrightarrow[N \to \infty]{} e^{-1} \approx 0.368.$$

Exactly k matches: choose the k people who match and require no match among the other $N-k$; this gives

$$P(\text{exactly } k) = \binom{N}{k} \frac{(N-k)!}{N!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} = \frac{1}{k!} \sum_{r=0}^{N-k} \frac{(-1)^r}{r!} \approx \frac{e^{-1}}{k!},$$

a Poisson(1) probability (Lecture 4).

Example 2.5 (Couples at a round table)

Ten couples sit at random around a round table (19! equally likely circular orders). Let E_i be the event that couple i sit together. Gluing r given couples into blocks leaves $20-r$ units, which can be arranged in $(19-r)!$ circular orders, and each block in 2 ways. Hence

$$P(\text{no couple together}) = \sum_{r=0}^{10} (-1)^r \binom{10}{r} \frac{2^r (19-r)!}{19!} \approx 0.3395.$$

2.4 Probability as a continuous set function

Motivation. Many events are limits of simpler ones, for example "eventually" or "infinitely often". Countable additivity implies that probability respects such monotone limits, which is what lets us compute probabilities of infinite events.

A sequence of events is **increasing** if $E_n \subset E_{n+1}$, with limit $\bigcup E_n$, and **decreasing** if $E_n \supset E_{n+1}$, with limit $\bigcap E_n$.

Proposition 2.3 (Continuity)

For an increasing or decreasing sequence, $\lim_n P(E_n) = P(\lim_n E_n)$.

Proof. Increasing case: put $F_1 = E_1$ and $F_n = E_n E_{n-1}^c$. The F_n are disjoint, $\bigcup_1^n F_i = E_n$ and $\bigcup_1^\infty F_i = \bigcup_1^\infty E_i$, so by axiom (iii)

$$P\Big(\bigcup_i E_i\Big) = \sum_{i=1}^{\infty} P(F_i) = \lim_n \sum_{i=1}^{n} P(F_i) = \lim_n P(E_n).$$

Decreasing case: apply this to the increasing E_n^c and use De Morgan. ■

Example 2.6

A fair coin is tossed forever. Let E_n be the event that the first n tosses are all tails; then E_n decreases to the event T that no head ever appears, so $P(T) = \lim 2^{-n} = 0$. Thus a head appears eventually with probability 1, although "all tails" is a possible outcome: probability zero does not mean impossible.

Subjective probability (a person's degree of belief) obeys the same axioms if beliefs are to be coherent, so the theory applies to it as well.

2.5 Selected exercises

Motivation. These are Ross's theoretical exercises for Chapter 2 (book numbering). They supply the induction proof of inclusion–exclusion for a general sample space, a Bonferroni bound, and two standard counting and modelling facts.

Exercise 2.8 (Counting partitions)

Let T_n be the number of partitions of $\{1,\dots,n\}$ into nonempty blocks, and put $T_0 = 1$.
(a) Show that $T_3 = 5$.
(b) Show that $T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k$ for $n \geq 1$, and deduce $T_4 = 15$.

Solution.

(a) The partitions of $\{1,2,3\}$ are $\{123\}$, $\{1\}\{23\}$, $\{2\}\{13\}$, $\{3\}\{12\}$ and $\{1\}\{2\}\{3\}$.

(b)

Step 1. Call the element $n+1$ special. A partition π of $\{1,\dots,n+1\}$ determines the block $B \ni n+1$ and the partition π' of the remaining set $R = \{1,\dots,n\} \setminus B$ formed by the other blocks (π' is empty if $R = \emptyset$). Conversely, a subset $A \subseteq \{1,\dots,n\}$ and a partition π' of $\{1,\dots,n\} \setminus A$ give back the partition $\{A \cup \{n+1\}\} \cup \pi'$. These maps are inverse to each other.

Step 2. Classify by $k = |R| \in \{0,\dots,n\}$. There are $\binom{n}{n-k} = \binom{n}{k}$ choices of A with $|A| = n-k$, and then T_k choices of π' when $k \geq 1$, and exactly one (the empty partition) when $k = 0$. Hence

$$T_{n+1} = 1 + \sum_{k=1}^{n} \binom{n}{k} T_k.$$

Step 3. With $T_1 = 1$, $T_2 = 2$ and $T_3 = 5$, $T_4 = 1 + 3T_1 + 3T_2 + T_3 = 1+3+6+5 = 15$. Continuing, $T_5 = 52$, $T_6 = 203$, …, $T_{10} = 115{,}975$. ∎

Exercise 2.14 (Inclusion–exclusion by induction)

Prove Theorem 2.2 for an arbitrary sample space by induction on n.

Solution. For $\emptyset \neq I \subseteq \{1,\dots,n\}$ write $E_I = \bigcap_{i \in I} E_i$. The claim is $P(\bigcup_{i=1}^{n} E_i) = \sum_I (-1)^{|I|+1} P(E_I)$, the sum over nonempty $I \subseteq \{1,\dots,n\}$.

Step 1. For $n = 1$ both sides equal $P(E_1)$; for $n = 2$ the claim is Proposition 2.1(c).

Step 2. Assume the claim for every family of n events and let $E_1,\dots,E_{n+1}$ be given. Put $U = \bigcup_{i=1}^{n} E_i$. By Proposition 2.1(c) and $UE_{n+1} = \bigcup_{i=1}^{n} (E_i E_{n+1})$,

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = P(U) + P(E_{n+1}) - P\Big(\bigcup_{i=1}^{n} E_i E_{n+1}\Big).$$

Step 3. Apply the induction hypothesis to $E_1,\dots,E_n$ and to $E_1E_{n+1},\dots,E_nE_{n+1}$. For $\emptyset \neq J \subseteq \{1,\dots,n\}$ the intersection of the events E_jE_{n+1}, $j \in J$, is $E_{J \cup \{n+1\}}$, so

$$P\Big(\bigcup_{i=1}^{n+1} E_i\Big) = \sum_J (-1)^{|J|+1} P(E_J) + P(E_{\{n+1\}}) - \sum_J (-1)^{|J|+1} P(E_{J \cup \{n+1\}}),$$

both sums over nonempty $J \subseteq \{1,\dots,n\}$.

Step 4. Every nonempty $I \subseteq \{1, \dots, n+1\}$ occurs exactly once on the right: if $n+1 \notin I$, in the first sum with sign $(-1)^{|I|+1}$; if $I = \{n+1\}$, as the middle term with sign $+1 = (-1)^{1+1}$; if $I = J \cup \{n+1\}$ with $J \neq \emptyset$, in the last sum with sign $-(-1)^{|J|+1} = (-1)^{|I|+1}$ since $|I| = |J| + 1$. This is the claim for $n+1$ events. ■

Exercise 2.16 (Bonferroni's inequality)

Show that for any events $E_1, \dots, E_n$,

$$P(E_1 E_2 \cdots E_n) \geq P(E_1) + \cdots + P(E_n) - (n-1).$$

Solution.

Step 1. ($n = 2$.) By Proposition 2.1(c) and $P(E \cup F) \leq 1$, $P(EF) = P(E) + P(F) - P(E \cup F) \geq P(E) + P(F) - 1$. For $n = 1$ the claim is an equality.

Step 2. (Induction.) If the claim holds for n events, apply Step 1 to $E = E_1 \cdots E_n$ and $F = E_{n+1}$:

$$\begin{aligned} P(E_1 \cdots E_{n+1}) &\geq P(E_1 \cdots E_n) + P(E_{n+1}) - 1 \\ &\geq \sum_{i=1}^{n} P(E_i) - (n-1) + P(E_{n+1}) - 1 = \sum_{i=1}^{n+1} P(E_i) - n. \end{aligned}$$

For example, if $P(E) = 0.9$ and $P(F) = 0.8$, then $P(EF) \geq 0.7$ whatever the dependence between E and F. ■

Exercise 2.19 (Waiting for the r-th red ball)

An urn contains n red and m blue balls. They are withdrawn one at a time, at random, until a total of $r \leq n$ red balls have been withdrawn. Find the probability that exactly k balls are withdrawn.

Solution.

Step 1. Label the balls and take as sample space the $(n+m)!$ equally likely orders in which all of them would be withdrawn. Exactly k balls are withdrawn iff the first $k-1$ positions contain exactly $r-1$ red balls and position k holds a red ball; this requires $r \leq k \leq r+m$.

Step 2. Count such orders: choose the red and blue balls in the first $k-1$ positions ($\binom{n}{r-1}\binom{m}{k-r}$ ways) and their order ($(k-1)!$ ways), the red ball in position k among the $n-r+1$ remaining red balls, and the order of the last $n+m-k$ balls ($(n+m-k)!$ ways).

Step 3. Dividing by $(n+m)!$ and using $\binom{n+m}{k-1}^{-1} = (k-1)!\,(n+m-k+1)!/(n+m)!$,

$$P(k \text{ balls withdrawn}) = \frac{\binom{n}{r-1}\binom{m}{k-r}}{\binom{n+m}{k-1}} \cdot \frac{n-r+1}{n+m-k+1}, \qquad r \leq k \leq r+m.$$

The first factor is the hypergeometric probability of $r-1$ red balls among the first $k-1$ draws, and the second the probability that the next draw is red given this. ■

Exercise 2.20 (No uniform distribution on a countable set)

Let the sample space $S = \{s_1, s_2, \dots\}$ be countably infinite. Show that not all outcomes can be equally likely. Can all outcomes have positive probability?

Solution.

Step 1. The singletons $\{s_i\}$ are mutually exclusive with union S, so by axioms (ii) and (iii), $1 = P(S) = \sum_{i=1}^{\infty} P(\{s_i\})$.

Step 2. If $P(\{s_i\}) = p$ for every i, the series is 0 when $p = 0$ and $+\infty$ when $p > 0$; neither equals 1. So the outcomes cannot all be equally likely.

Step 3. All outcomes can have positive probability: $P(\{s_i\}) = 2^{-i}$, extended to every event E by $P(E) = \sum_{i:s_i \in E} 2^{-i}$, is nonnegative, has $P(S) = \sum_i 2^{-i} = 1$, and is countably additive because a series of nonnegative terms may be summed in any grouping. ■